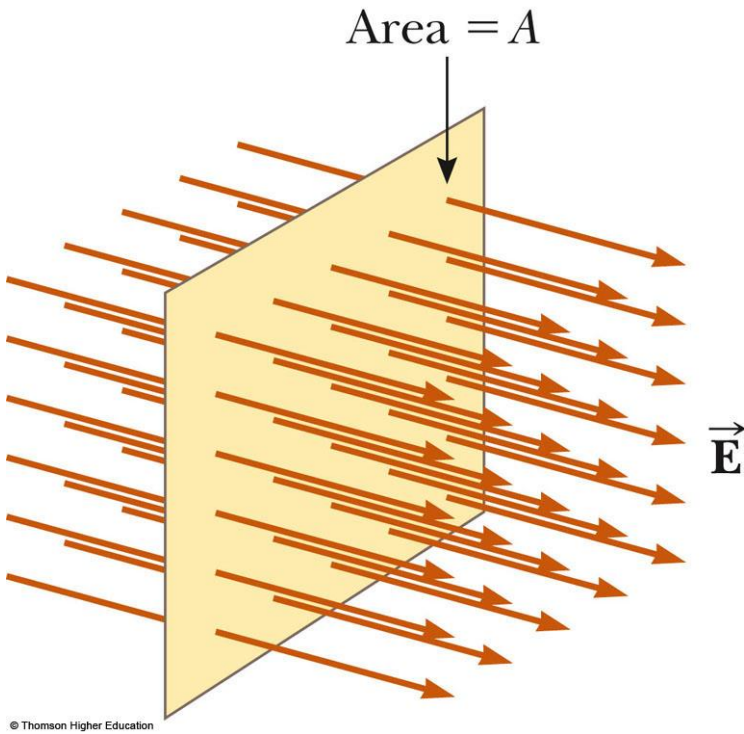
Chapter 24
Gauss's Law

Sec 4.1: Electric Flux

- ***Electric flux*** is the product of the magnitude of the electric field and the surface area, A, perpendicular to the field
- $\Phi_E = EA$
- The electric flux is proportional to the number of electric field lines penetrating some surface
- The flux is a maximum when the surface is perpendicular to the field
- The flux is zero when the surface is parallel to the field
- If the field varies over the surface, $\Phi = EA \cos \theta$ is valid for only a small element of the area

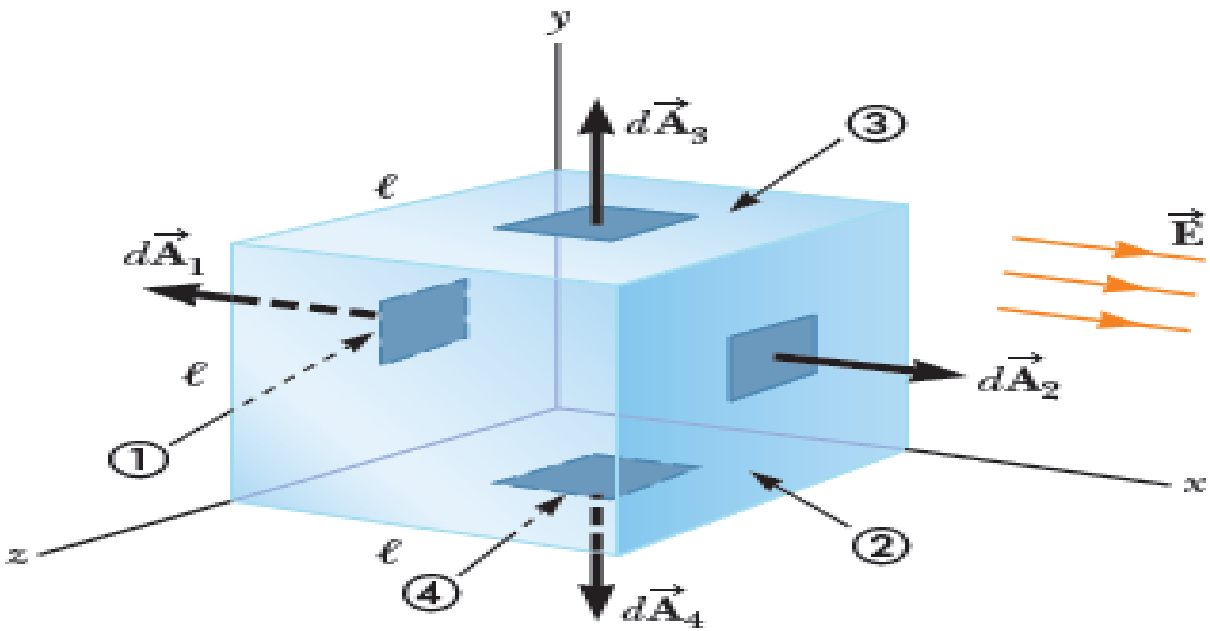


- The units of electric flux will be $N \cdot m^2/C$
- The ***net*** flux through the surface is proportional to the net number of lines leaving the surface
  - This net number of lines is the number of lines leaving the surface minus the number entering the surface

Quick Quiz 24.1 Suppose a point charge is located at the center of a spherical surface. The electric field at the surface of the sphere and the total flux through the sphere are determined. Now the radius of the sphere is halved. What happens to the flux through the sphere and the magnitude of the electric field at the surface of the sphere? **(a)** The flux and field both increase. **(b)** The flux and field both decrease. **(c)** The flux increases, and the field decreases. **(d)** The flux decreases, and the field increases. **(e)** The flux remains the same, and the field increases. **(f)** The flux decreases, and the field remains the same.

Example 24.1 Flux Through a Cube

Consider a uniform electric field $\vec{\mathbf{E}}$ oriented in the x direction in empty space. A cube of edge length ℓ is placed in the field, oriented as shown in Figure 24.5. Find the net electric flux through the surface of the cube.



$$\vec{\mathbf{E}} \cdot \vec{\mathbf{A}} = E(\cos 180°)A = -EA = -EA = -E\ell^2$$

$$\vec{\mathbf{E}} \cdot \vec{\mathbf{A}} = E(\cos 0°)A = EA = +EA = E\ell^2$$

$$\Phi_E = -E\ell^2 + E\ell^2 + 0 + 0 + 0 + 0 = \boxed{0}$$

Sec 24.2: Gauss's Law

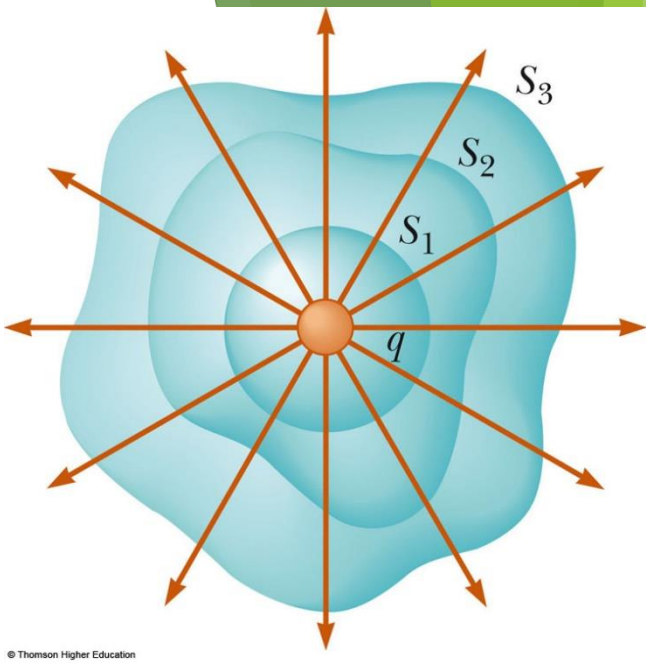
- Gauss's law is an expression of the general relationship between the net electric flux through a closed surface and the charge enclosed by the surface
  - The closed surface is often called a *gaussian surface*
- Gauss's law is of fundamental importance in the study of electric fields
- The field lines are directed radially outward and are perpendicular to the surface at every point
- $\Phi = EA \cos \theta$
- This will be the net flux through the gaussian surface,

$$\Phi_E = 4\pi k_e q = \frac{q}{\varepsilon_o}$$

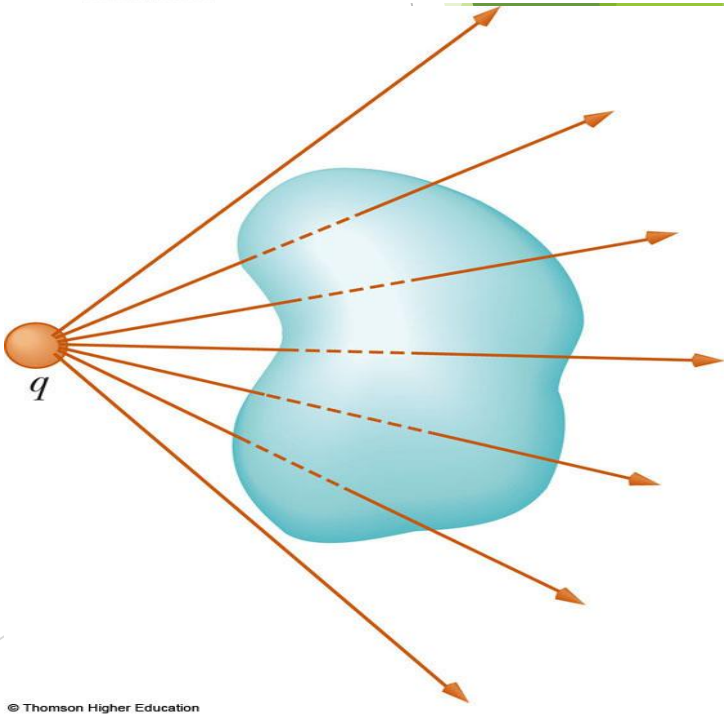
- The net flux through any closed surface surrounding a point charge, q, is given by q/ε_o and is independent of the shape of that surface
- The net electric flux through a closed surface that surrounds no charge is zero
- Gauss's law states $$\Phi_E = \vec{\mathbf{E}} \cdot \vec{\mathbf{A}} = \frac{q_{in}}{\varepsilon_o}$$
  - q_{in} is the net charge inside the surface
- $\vec{\mathbf{E}}$ represents the electric field at any point on the surface
  - $\vec{\mathbf{E}}$ is the *total electric field* and may have contributions from charges both inside and outside of the surface
- Although Gauss's law can, in theory, be solved to find $\vec{\mathbf{E}}$ for any charge configuration, in practice it is limited to symmetric situations

Gaussian Surface, Example

- Closed surfaces of various shapes can surround the charge (same)
  - Only S_1 is spherical
- Verifies the net flux through any closed surface surrounding a point charge q is given by q/ε_0 and is independent of the shape of the surface



- The charge is outside the closed surface with an arbitrary shape
- Any field line entering the surface leaves at another point
- Verifies the electric flux through a closed surface that surrounds no charge is zero



Quick Quiz 24.2 If the net flux through a gaussian surface is *zero*, the following four statements *could be true*. Which of the statements *must be true*? **(a)** There are no charges inside the surface. **(b)** The net charge inside the surface is zero. **(c)** The electric field is zero everywhere on the surface. **(d)** The number of electric field lines entering the surface equals the number leaving the surface.

Conceptual Example 24.2 Flux Due to a Point Charge

A spherical gaussian surface surrounds a point charge q. Describe what happens to the total flux through the surface if (A) the charge is tripled, (B) the radius of the sphere is doubled, (C) the surface is changed to a cube, and (D) the charge is moved to another location inside the surface.

SOLUTION

(A) The flux through the surface is tripled because flux is proportional to the amount of charge inside the surface.

(B) The flux does not change because all electric field lines from the charge pass through the sphere, regardless of its radius.

(C) The flux does not change when the shape of the gaussian surface changes because all electric field lines from the charge pass through the surface, regardless of its shape.

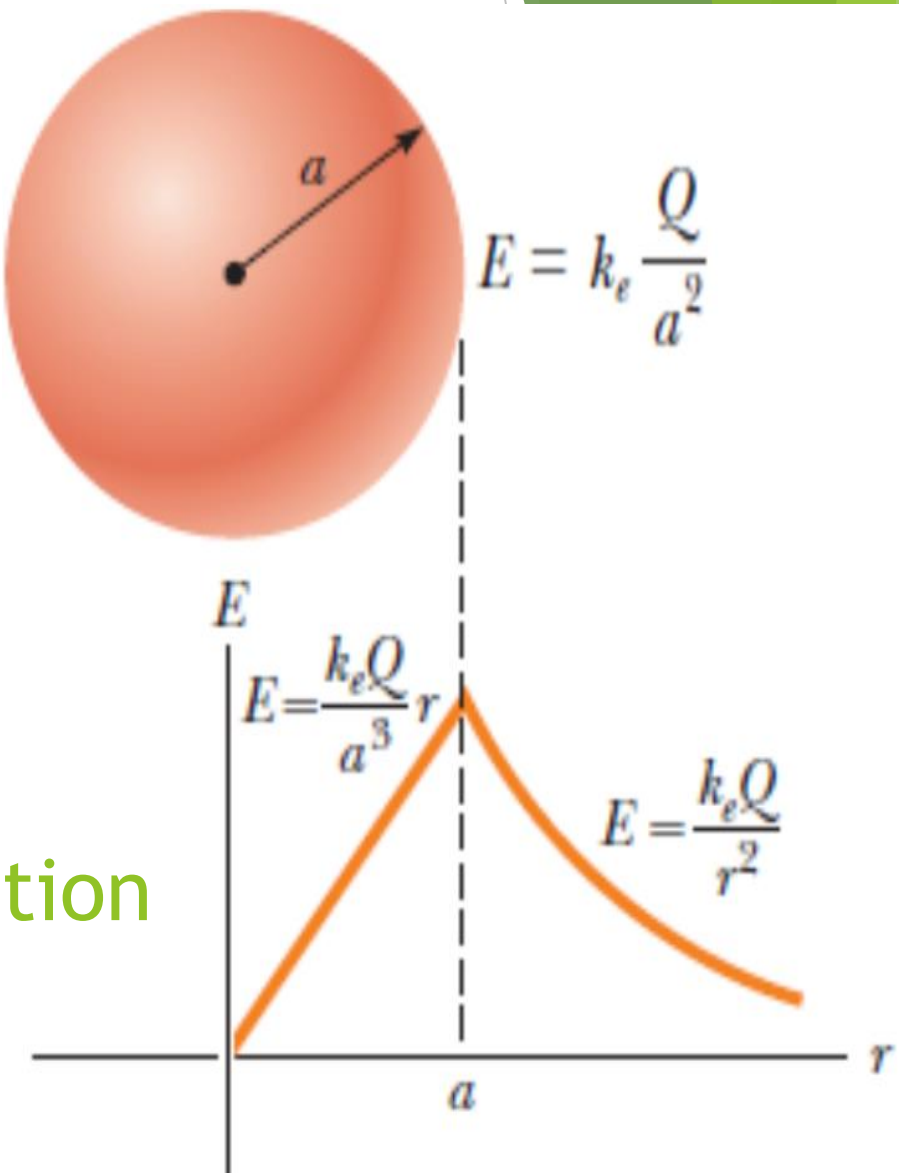
(D) The flux does not change when the charge is moved to another location inside that surface because Gauss's law refers to the total charge enclosed, regardless of where the charge is located inside the surface.

Sec 24.3: Application of Gauss's Law to Various Charge Distributions

- To use Gauss's law, you want to choose a gaussian surface over which the surface integral can be simplified and the electric field determined

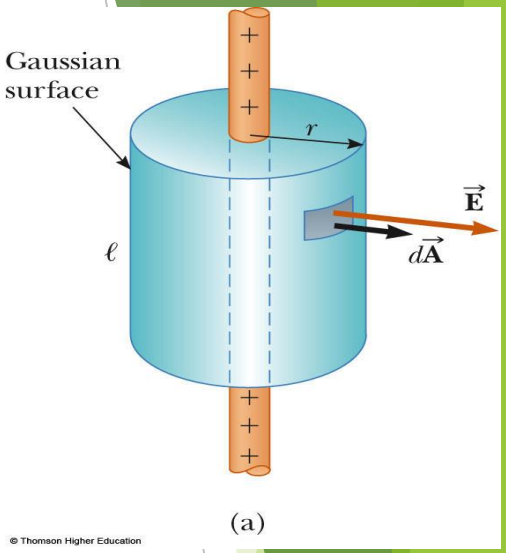
- Remember, the gaussian surface is a surface you choose, it does not have to coincide with a real surface

Field Due to a Spherically Symmetric Charge Distribution



Field at a Distance from a Line of Charge

$$E = \frac{\lambda}{2\pi\varepsilon_o r} = 2k_e \frac{\lambda}{r}$$

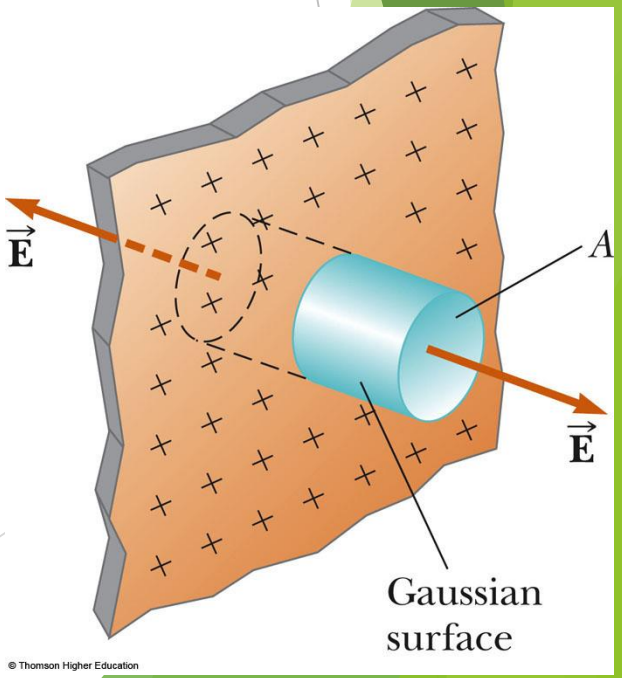


Field Due to a Plane of Charge

$$E = \frac{\sigma}{2\varepsilon_o}$$

- For two plates will be :

$$E = \frac{\sigma}{\varepsilon_o}$$



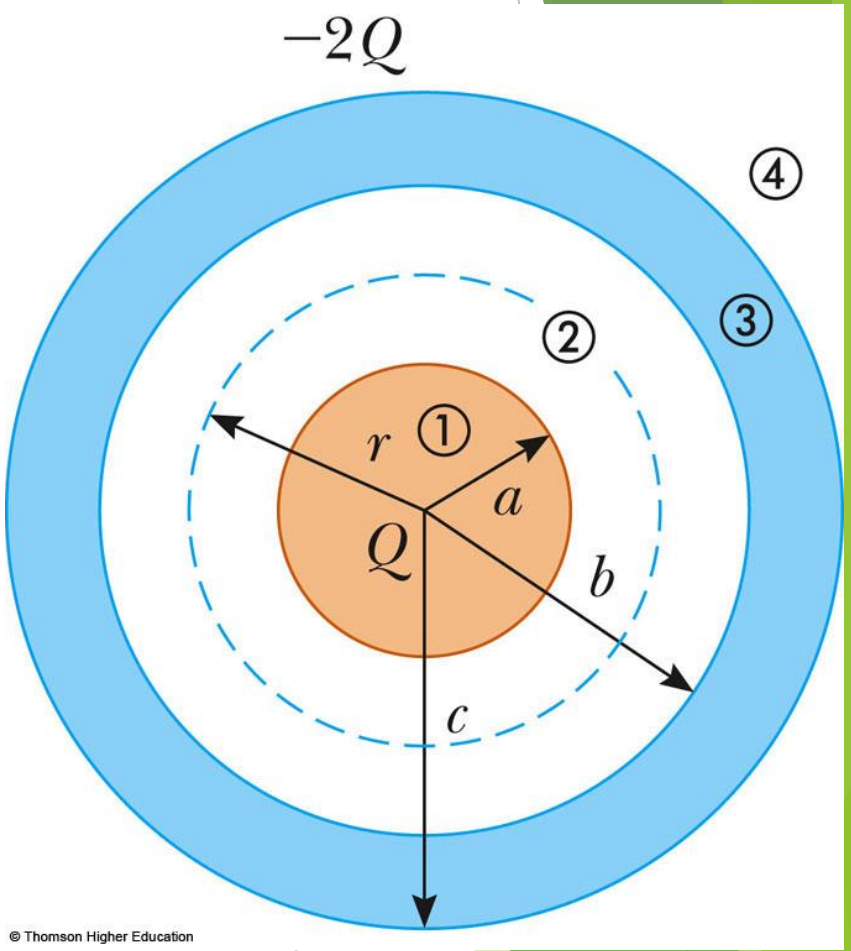
Sec 24.4: Conductors in Electrostatic Equilibrium

- When there is no net motion of charge within a conductor, the conductor is said to be in **electrostatic equilibrium**
- The electric field is zero everywhere inside the conductor, Whether the conductor is solid or hollow
- If an isolated conductor carries a charge, the charge resides on its surface
- The electric field just outside a charged conductor is perpendicular to the surface and has a magnitude of σ/ε_o
- On an irregularly shaped conductor, the surface charge density is greatest at locations where the radius of curvature is the smallest

Sphere and Shell Example

$$E_1 = k_e \frac{Q}{a^3} r \quad (for\ r < a)$$

$$E_2 = k_e \frac{Q}{r^2} \quad (for\ a < r < b)$$

$$E_3 = 0 \quad (for\ b < r < c)$$

$$E_4 = -k_e \frac{Q}{r^2} \quad (for\ r > c)$$